BIOCHEMISTRY ACADEMY

Faculty of Medicine
Ain Shams University
2026 - 2027

RESPIRATORY MODULE

CHAPTER

Glucose Oxidation & Shuttles

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Cellular Respiration

- **Using O_2 and energy-derived from oxidizing fuel** (CHO, lipids & protein) to generate **ATP**.
- Cells require O_2 to generate ATP from the oxidation of fuels to CO_2 and water.
- Cellular respiration uses **over 90% of the O_2 we inhale.**

Phases of Cellular Respiration

Phase 1
Oxidation of fuels (carbohydrates, lipid & protein)

- 0xidation of (glucose, fatty acids, and amino acids) → **acetyl CoA.**
- Complete oxidation of the acetyl group to CO_2 occurs in the tricarboxylic acid (TCA) cycle, which **collects the energy as NADH and $FADH_2$.**

Phase 2
ATP synthesis

- Electrons are transferred from **NADH and $FADH_2$ to O2 by the electron transport chain (ETC)** Called **oxidative phosphorylation.**
- These series of reactions are coupled with the synthesis of **ATP.**

Phase 3
Consumption of ATP

ATP are used for processes such as:
- ✓ muscle contraction (**mechanical**)
- ✓ Na^+-K^+ pump (**transport**),
- ✓ Synthesis of larger molecules such as DNA in anabolic pathways (**biosynthetic**).
- ✓ Or detoxification (**biochemical**).

ATP is hydrolyzed to ADP + (Pi), or AMP + pyrophosphate (PPi).

- **Mitochondria** is the site for cellular respiration:-
  - ✓ TCA cycle "except succinate dehydrogenase" and oxidation of fatty acids located in the **mitochondrial matrix (phase 1).**
  - ✓ ETC are in the **Inner mitochondrial membrane (phase 2).**

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Examples of cellular respiration

- **Glucose** is a **universal fuel** used to generate ATP in every cell type in the body.
- **In glycolysis,** 1 glucose is oxidized to 2 pyruvate + 2 NADH by cytosolic enzymes + 2 ATP by substrate-level phosphorylation.

1. In anaerobic glycolysis

- NADH is reoxidized by conversion of pyruvate to **lactate**, enters the blood.
- Produce only **2 ATP.**
- Important for cells **lack mitochondria** as 'RBCS & kidney medulla' or tissues with **diminished blood flow** (exercising muscle).

2. In aerobic glycolysis

- NADH produced from glycolysis is reoxidized via the **electron transport chain, and the pyruvate is transported into the mitochondria** where it is oxidized to acetyl CoA which is the major fuel of the TCA cycle.

Fate of the end products of aerobic glycolysis (pyruvate and NADH)

A. Cytosolic NADH:

- Cytosolic NADH **cannot directly pass through inner mitochondrial membrane** to reach the electron transport chain.
- NADH is indirectly transferred to the inside of the mitochondria by what is called the shuttle mechanisms as **malate aspartate shuttle or Glycerol phosphate shuttle.**

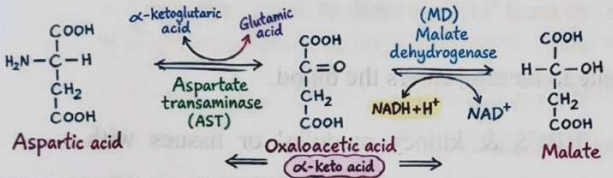
B. Pyruvate:

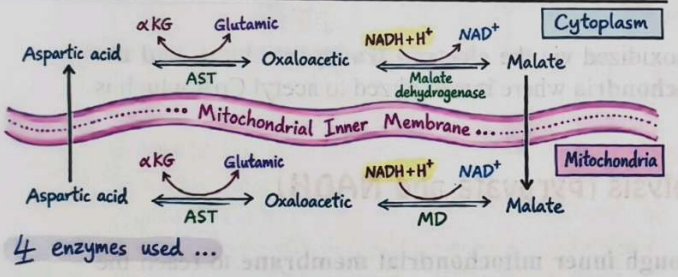
- In presence of O_2 and mitochondria, pyruvate is transported to mitochondria and undergoes **oxidative decarboxylation to acetyl CoA**

Shuttles of transfer NADH to Mitochondria

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1. Malate shuttle (Malate-Aspartate)

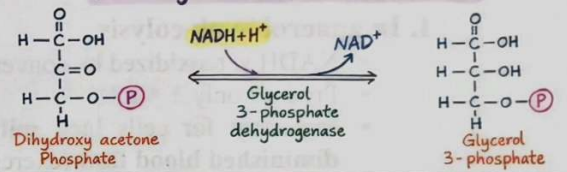


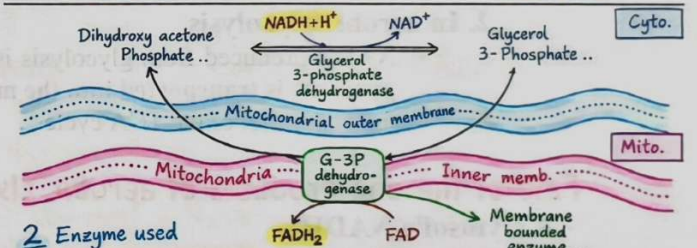


- Malate is transported to the mitochondria using **malate translocase**
- Oxaloacetate **can't** cross inner mitochondrial membrane.
- Aspartate is transported out to the cytosol using an **aspartate/glutamate exchange translocase.**
- NADH produce **3 ATP.**

2. Glycerol phosphate shuttle

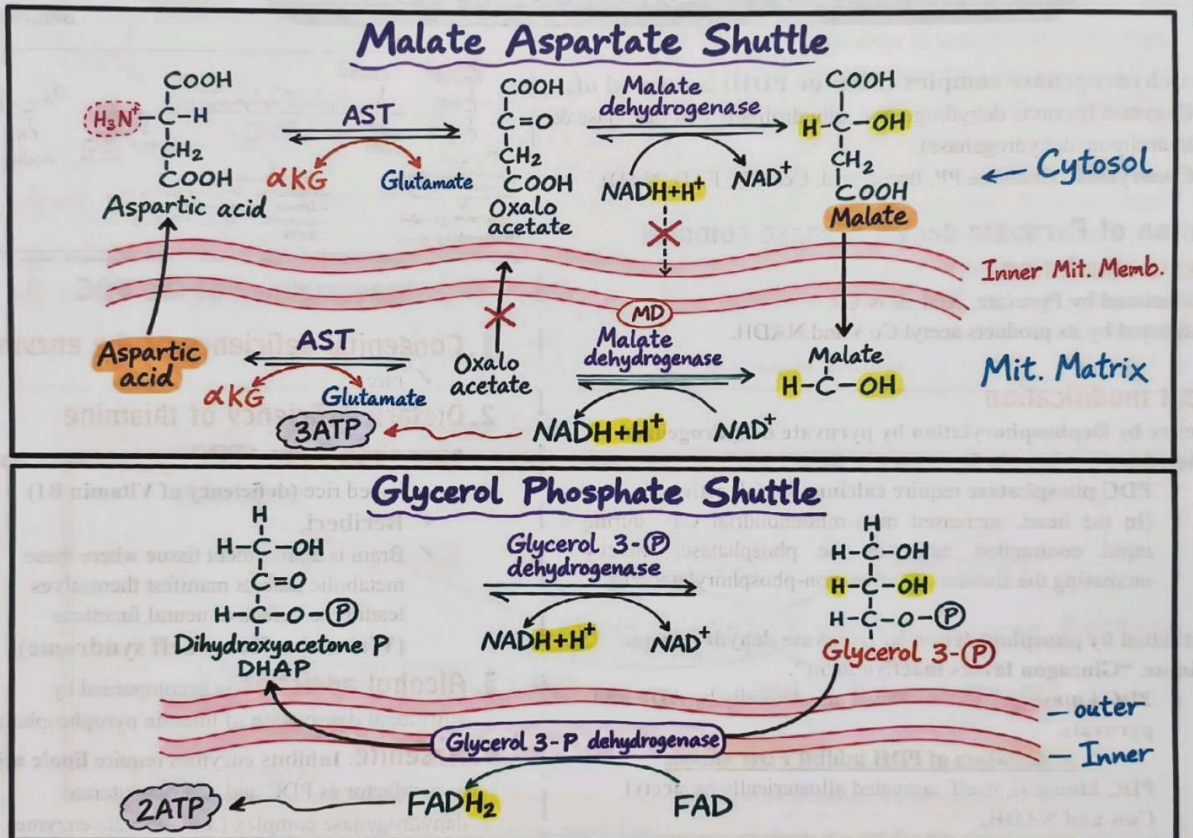
The major shuttle in most tissues





- Glycerol-3P **diffuse** to the inner mitochondrial membrane.
- Electrons are donated to FAD forming **$FADH_2$** → produce **2 ATP.**
- **DHAP return to cytosol.**

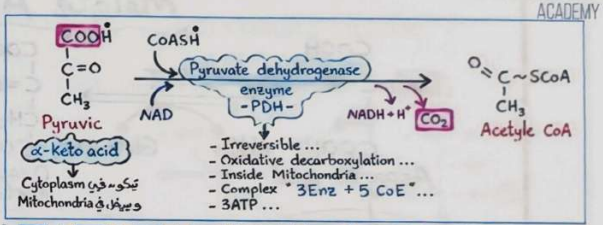
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Oxidative decarboxylation of Pyruvate

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- **pyruvate dehydrogenase complex (PDC or PDH) is formed of:**
  - ✓ **3 Enzymes** (pyruvic dehydrogenase, dihydrolipoic transacetylase & dihydrolipoic dehydrogenase).
  - ✓ **5 Coenzymes** (Thiamine PP, lipoic acid, CoASH, FAD, NAD).



Regulation of Pyruvate dehydrogenase complex

1. Allosteric regulation

- **Stimulated** by Pyruvate, ADP & NAD.
- **Inhibited** by its products acetyl CoA and NADH.

2. Covalent modification

- **Active by Dephosphorylation by pyruvate dehydrogenase phosphatase. "Insulin favors active form".**
  - ✓ **PDC phosphatase** require **calcium** for full activity.
  - ✓ (In the heart, increased intra-mitochondrial Ca^{2+} during rapid contraction activates the phosphatase, thereby increasing the amount of active, non-phosphorylated PDC).
- **Inhibited by phosphorylation by pyruvate dehydrogenase Kinase. "Glucagon favors inactive form".**
  - ✓ **PDC kinase** is, itself, inhibited allosterically by **ADP and pyruvate.**

    Stimulators of PDH inhibit PDH kinase.
  - ✓ **PDC kinase** is, itself, activated allosterically by **acetyl CoA and NADH.**

    Inhibitors of PDH activate PDH kinase.

Abnormalities of the PDC

1. Congenital deficiency of the enzyme

- ✓ rare

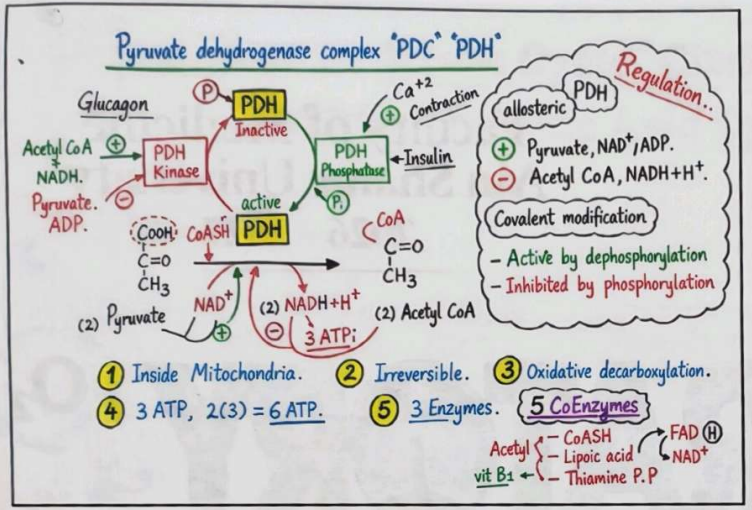
2. Dietary deficiency of thiamine pyrophosphate (TPP) in populations that use polished rice (deficiency of Vitamin B1)

- ✓ **Beriberi.**
- ✓ Brain is a prominent tissue where these metabolic defects manifest themselves leading to defective neural functions **(Wernicke-Korsakoff syndrome).**

3. Alcohol addiction is accompanied by nutritional deprivation of thiamin pyrophosphate.

4. Arsenite: **Inhibits** enzymes require **lipoic acid** as a cofactor as PDC and α- ketoglutarate dehydrogenase complex (a TCA cycle enzyme).

- ✓ **Pyruvate & lactate accumulate.**

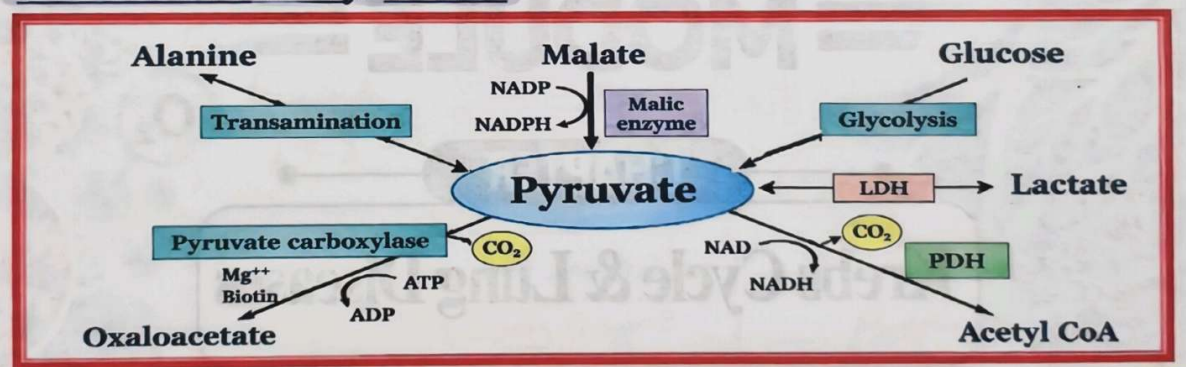


Important Notes:

- **Phase 2 of cellular respiration:**
  - ✓ The energy derived from fuel oxidation is converted to the high-energy phosphate bonds or ATP by the process or **oxidative phosphorylation.**
- Some individuals with **alcoholism** develop **Wernicke-Korsakoff syndrome** because brain depends on glucose as a preferred fuel, so it is a prominent tissue where these metabolic defects manifest themselves.
- **Abnormalities** in PDC leads to **increase lactic acid** in blood and decrease pH, this serious condition called **lactic acidosis.**

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Other Fates of Pyruvate:



***Pyruvate carboxylase* and *PDC* catalyze *irreversible* reactions. While *LDH* and *transamination* reactions are reversible.**

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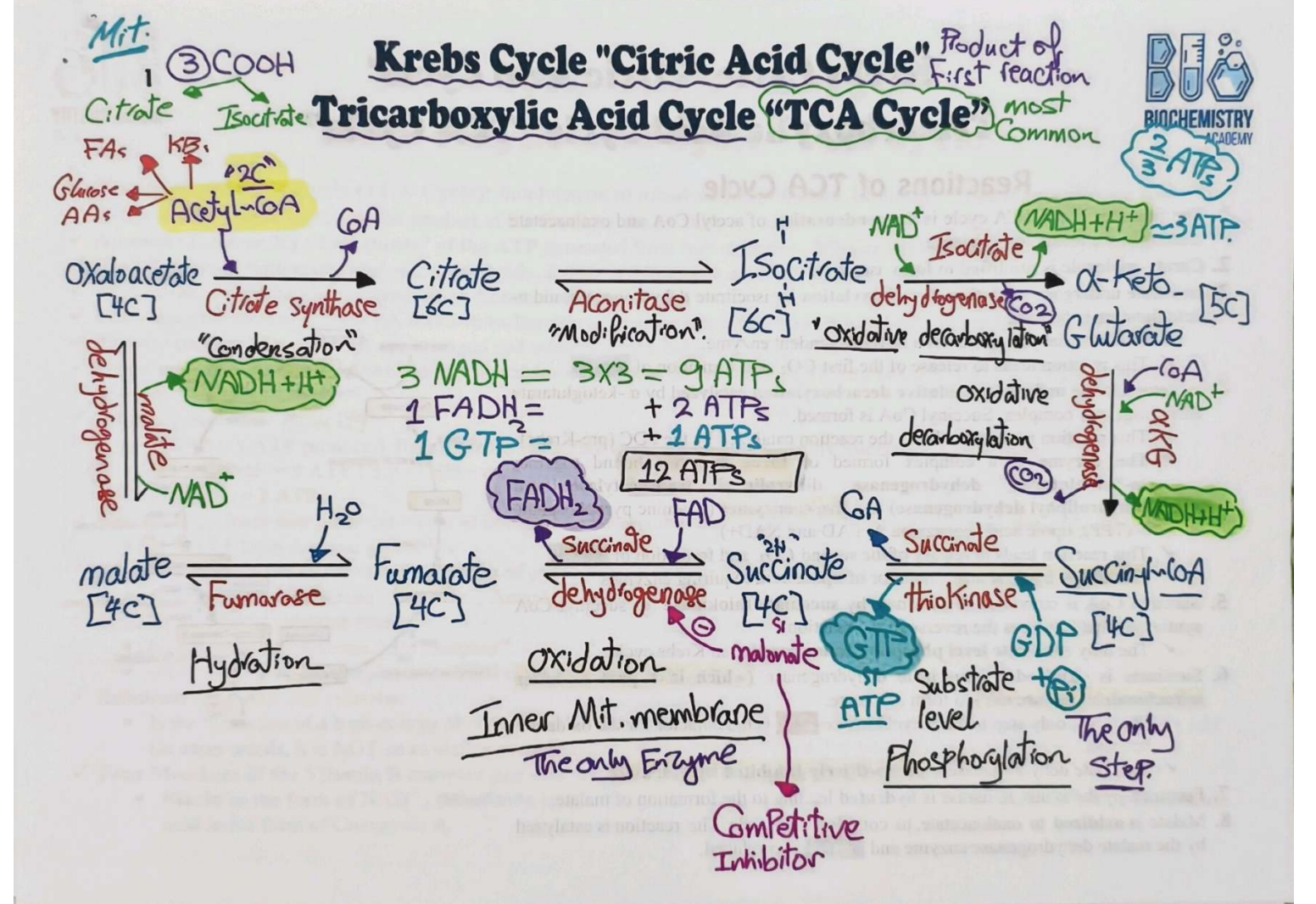
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RESPIRATORY MODULE

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Krebs Cycle & Lung Diseases

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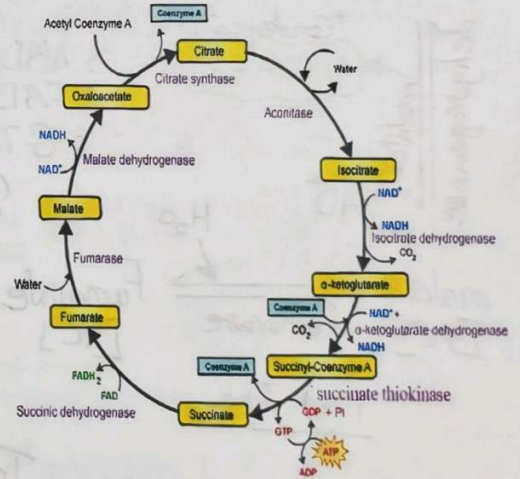

Mit.
Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"
Product of First reaction
most Common
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2/3 ATPs
③COOH
Citrate
Isocitrate
FAs
KBs
Glucose
AAs
"2C"
Acetyl-CoA
CoA
Oxaloacetate [4C]
Citrate synthase
Citrate [6C]
Aconitase
"Modification"
Isocitrate [6C]
NAD+
NADH+H+
≈3ATP
Isocitrate dehydrogenase
CO2
"oxidative decarboxylation"
α-keto-Glutarate [5C]
"Condensation"
NADH+H+
dehydrogenase
malate
NAD+
3 NADH = 3X3 = 9 ATPs
1 FADH2 = + 2 ATPs
1 GTP = + 1 ATPs
12 ATPs
oxidative decarboxylation
CO2
CoA
NAD+
αKG
dehydrogenase
NADH+H+
H2O
malate [4C]
Fumarase
Fumarate [4C]
Hydration
FADH2
FAD
Succinate dehydrogenase
Succinate [4C]
"2H"
CoA
Succinate
Succinyl-CoA [4C]
thioKinase
GDP
GTP
ATP
Substrate level Phosphorylation
The only step.
oxidation
malonate
Inner Mit. membrane
The only Enzyme
Competitive Inhibitor

Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"

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Reactions of TCA Cycle

1. The first step in the TCA cycle is the **condensation** of acetyl CoA and oxaloacetate leading to the formation of citrate.
2. Citrate molecule is **modified** to form isocitrate.
3. Isocitrate undergoes **oxidation decarboxylation** by isocitrate dehydrogenase and α-ketoglutarate is formed.
   - ✓ isocitrate dehydrogenase is a NAD-dependent enzyme.
   - ✓ This reaction leads to release of the first CO_2 and formation of **NADH**.
4. α -ketoglutarate undergoes **oxidative decarboxylation** catalyzed by α -ketoglutarate dehydrogenase complex. Succinyl CoA is formed.
   - ✓ This reaction is very similar to the reaction catalyzed by the PDC (pre-Krebs).
   - ✓ The enzyme is a complex formed of **three enzymes** bound together **(α-ketoglutarate dehydrogenase, dihyrolipoyl transacetylase and dihydrolipoyl dehydrogenase)** and **five coenzymes** (thiamine pyrophosphate = (TPP), lipoic acid , coenzyme A, FAD and NAD+).
   - ✓ This reaction leads to release of the second CO_2. and formation of **NADH**.
   - ✓ **"Inhibited by Arsenite** "inhibitor of **lipoic acid** requiring enzymes"
5. Succinyl CoA is converted to succinate by **succinate thiokinase** (or succinyl CoA synthetase that catalyzes the reverse of the reaction).
   - ✓ The only **substrate level phosphorylation** reaction in Krebs cycle.
6. Succinate is **oxidized** by succinate dehydrogenase **(which is a part of inner mitochondrial membrane)** to form Fumarate.
   - ✓ The is the only step in TCA cycle where **FAD** is the cofactor for the oxidation reaction.
   - ✓ Succinate dehydrogenase is **competitively Inhibited by Malonate.**
7. Fumarate by the action fumarase is **hydrated** leading to the formation of malate.
8. Malate is **oxidized** to oxaloacetate, to complete the cycle. The reaction is catalyzed by the malate dehydrogenase enzyme and **NADH** is produced.



Krebs Cycle "Citric Acid Cycle"
Tricarboxylic Acid Cycle "TCA Cycle"

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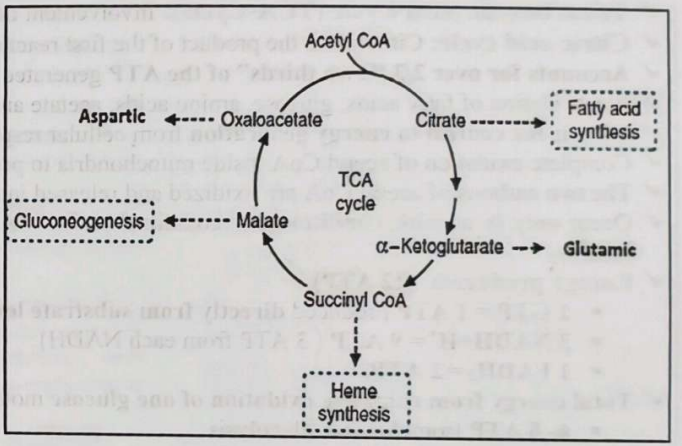
- ✓ **Tricarboxylic Acid Cycle (TCA-Cycle):** Involvement of tricarboxylates "Citrate, Isocitrate" **"The most common name".**
- ✓ **Citric acid cycle:** Citrate was the product of the first reaction of TCA.
- ✓ **Accounts for over 2/3 "Two thirds" of the ATP** generated from fuel oxidation. **"Occur exclusively in Mitochondria".**
- ✓ For oxidation of fatty acids, glucose, amino acids, acetate and ketone bodies "all generate acetyl CoA".
- ✓ **TCA is the central to energy generation** from cellular respiration.
- ✓ Complete oxidation of acetyl CoA inside mitochondria to produce **two CO_2, three $NADH+H^+$, one $FADH_2$ & one GTP.**
- ✓ The two carbons of acetyl CoA are oxidized and released in the form of **2 CO_2**
- ✓ Occur **only in aerobic conditions** to reoxidized back 3 $NADH+H^+$ & one $FADH_2$ into NAD & FAD by respiratory chain that need O_2.
- ✓ **Energy produced: (12 ATP)**
  - **1 GTP = 1 ATP** produced **directly from substrate level phosphorylation by Succinate Thiokinase.**
  - **3 $NADH+H^+$ = 9 ATP** (3 ATP from each NADH).
  - **1 $FADH_2$ = 2 ATP.**
- ✓ **Total energy from complete oxidation of one glucose** molecule under **aerobic conditions** will yield **a net of 36- 38 ATP**
  - **6- 8 ATP** from Aerobic glycolysis
  - **6 ATP** form Oxidative decarboxylation of pyruvate to acetyl CoA "PDH complex enzyme"
  - **(12 X 2) =24 ATP** Krebs' Cycle for 2 Acetyl CoA .
- ✓ **Oxidative decarboxylation reaction:**
  - Isocitrate dehydrogenase **"not complex".**
  - α ketoglutarate dehydrogenase (as PDH complex) need 3 Enzymes & 5 coenzymes.
- ✓ **Substrate level phosphorylation:**
  - Is the formation of a high-energy phosphate bond where none previously existed without the use of molecular O_2 or the ETC (in other words, it is NOT an oxidative phosphorylation). Occur only by **Succinate thiokinase in TCA Cycle.**
- ✓ **Four Members of the Vitamin B complex pay role in TCA Cycle "in the form of Coenzymes"**
  - **Niacin** in the form of NAD^+ , **Riboflavin** in the form of FAD, **Thiamin** (Vitamin B1) in the form of TPP, **Pantothenic acid** in the form of Coenzyme A.

Amphibolic aspects of TCA cycle

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Definition:

- TCA can act as **catabolic or anabolic pathway.**



فكرني أقولك حاجتين مهمين جدا

1- Catabolic role:

- ✓ TCA is **the final common pathway** for oxidation of CHO , lipids & protein.

2- Anabolic role:

a) Synthesis of fatty acids:

- ✓ **After high CHO meal,** Acetyl CoA is converted to **Citrate**, efflux to cytoplasm & hydrolyzed to give acetyl CoA in the cytoplasm for synthesis of Fatty acids & Lipogenesis.

b) Non-essential amino acids:

- ✓ By transamination
  - **Oxaloacetate ↔ aspartate.**
  - **α-ketoglutaric ↔ glutamate.**

c) Synthesis of heme in liver and bone marrow:

- **Succinyl CoA** + glycine → δ amino levulinic acids.

d) Gluconeogenesis "glucose synthesis":

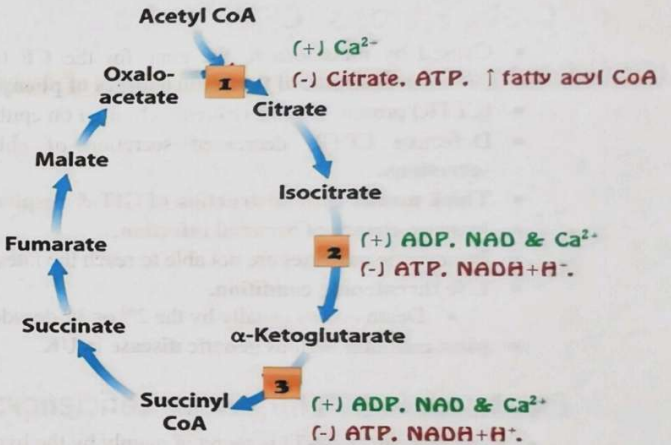
- **During fasting, Malate efflux** to cytoplasm to give glucose.

Regulation of Krebs' Cycle

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1-Allosteric regulation: "Fine"

1. Citrate synthase
2. Isocitrate dehydrogenase
3. α-ketoglutaric dehydrogenase".



2-Coarse control:

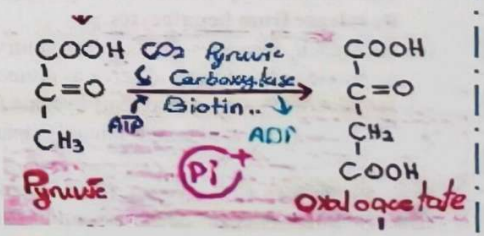
- The main function of the TCA cycle is energy production, which depends mainly on electron transport chain.
- **Availability of oxygen and the concentration of ADP** in the cell constitute the coarse control of the cycle. So **the rate of utilization of ATP** determines both the rate of respiration and the activity of the TCA cycle.

Anaplerotic Reactions:

Pathways or reactions that **replenish the intermediates** of the TCA cycle are referred to as anaplerotic reactions **("filling up").**

Pyruvate carboxylase is a major anaplerotic enzyme:

- ✓ Found in liver, Brain, adipocytes and Fibroblasts
- ✓ Pyruvate carboxylase is **activated by acetyl CoA** (When acetyl CoA accumulates. it allosterically stimulates pyruvate carboxylase to ensure proper concentration of oxaloacetate for the oxidation of the excess acetyl CoA).
- ✓



N.B.: **Four members of the vitamins B complex** used as coenzymes in Krebs' cycle:

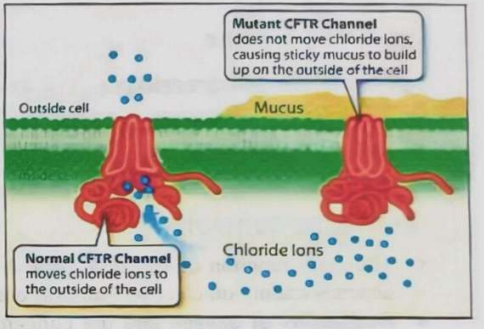
- **Vitamin B1 "Thiamine", Niacin, Riboflavin & Pantothenic acid.**

Molecular Basis of Lung Diseases

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Cystic Fibrosis (CF):

- Caused by mutations to the gene for the **CF transmembrane regulator (CFTR)** **"deletion of three nucleotides of phenylalanine".**
- (CFTR) protein act as a **chloride channel** on epithelium of Pancreas, lungs.
- Defective CFTR, decreased secretion of chloride → **thick, viscous secretions.**
- **Thick mucus with obstruction of GIT & respiratory tract.**
- Increase chance of bacterial **infection**.
- **Pancreatic enzymes** are not able to reach the intestine. **"Digestive problem"**
- **Life threatening condition.**
  - ✓ Death occurs usually by the 2nd or 3rd decades.
- Most common Serious **genetic disease** in **UK**.



Emphysema α1 antitrypsin deficiency:

- α_1-antitrypsin (α_1,-AT) is secreted mainly by the liver, monocyte & alveolar macrophage.
- It **inhibits a number of proteolytic enzymes (elastase)** that is **secreted by the neutrophils** and degrades elastin of alveolar walls if the lungs.
- The **α1 antitrypsin deficiency** is a genetic defect **alters the configuration of the alpha1-antitrypsin molecule and prevents its release from hepatocytes.**
- As a result, serum levels of alpha1-antitrypsin are decreased, leading to low alveolar concentrations, where the alpha1-antitrypsin molecule normally would serve as protection against proteases such neutrophils elastase. The excess elastase in alveoli **destroys alveolar walls and causes emphysema (no elastic recoil).**
- The accumulation of excess mutated alpha1-antitrypsin in hepatocytes can also lead to destruction of these cells and **liver disease**.

N.B.: Smoking adds to the severity of the disease. Smoking causes **oxidation and subsequent inactivation of methionine** in α1-antitrypsin (**Methionine 358** in a1- antitrypsin is required for the binding of a1- antitrypsin to its target elastase). So, **α1- antitrypsin can't bind to elastase.** This leads to lung **emphysema**.